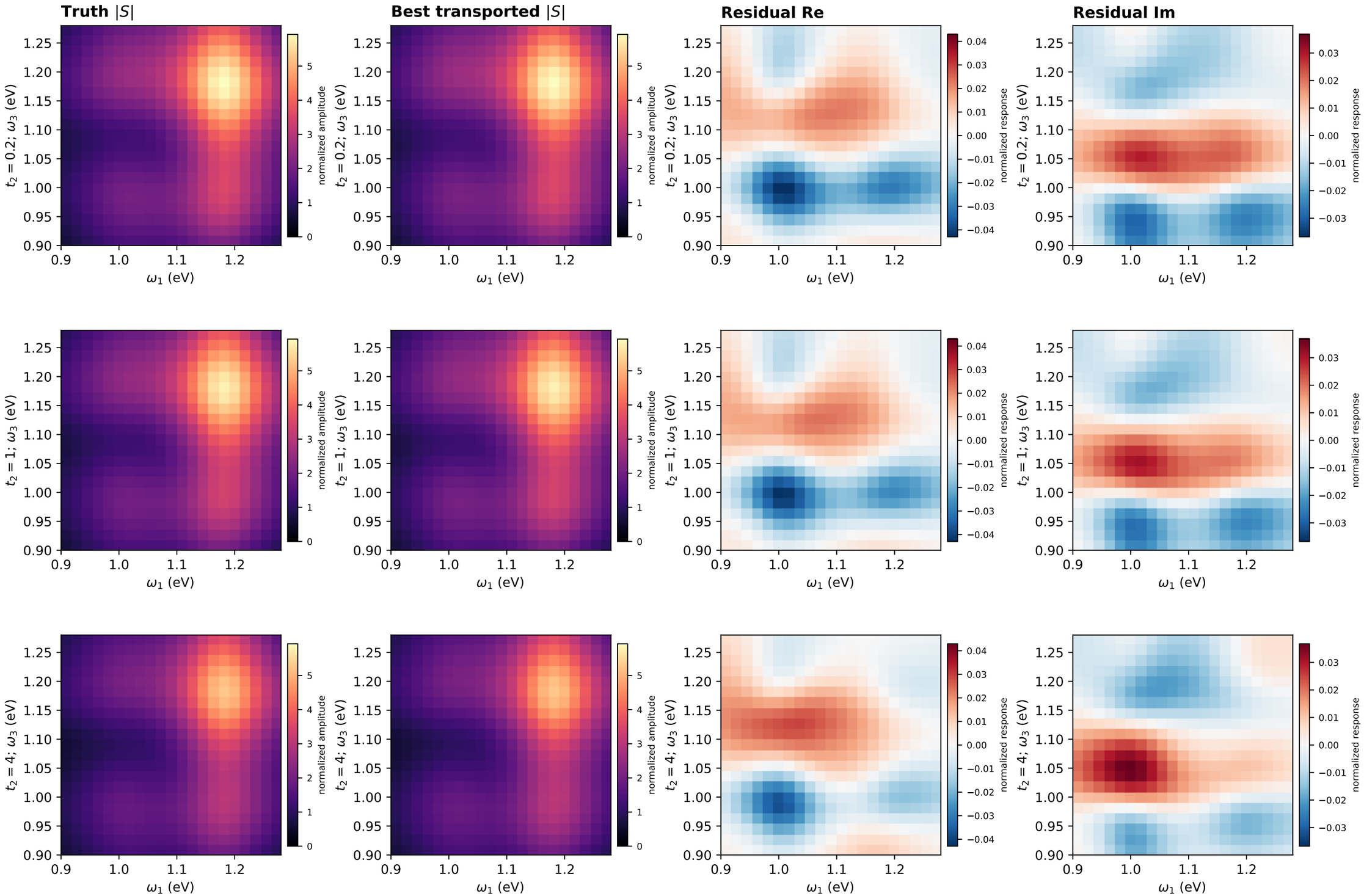
Dark-state truth and best sink-free Duhamel-transport fit

Truth: $k_d = 0.003$**; joint transport fit:** $J_{12} = 0.02558$**,** $k_{12} = 0.0124$**,** $\gamma_\phi = 0.02256$



Truth is the full RP+NRP response of the terminal-sink model. The candidate has no dark state. A rank-reduced reference pathway basis is weighted to preserve the full canonical sum at the reference; transport is third-order Duhamel/autodiff. All panels share a single truth-derived L2 normalization.